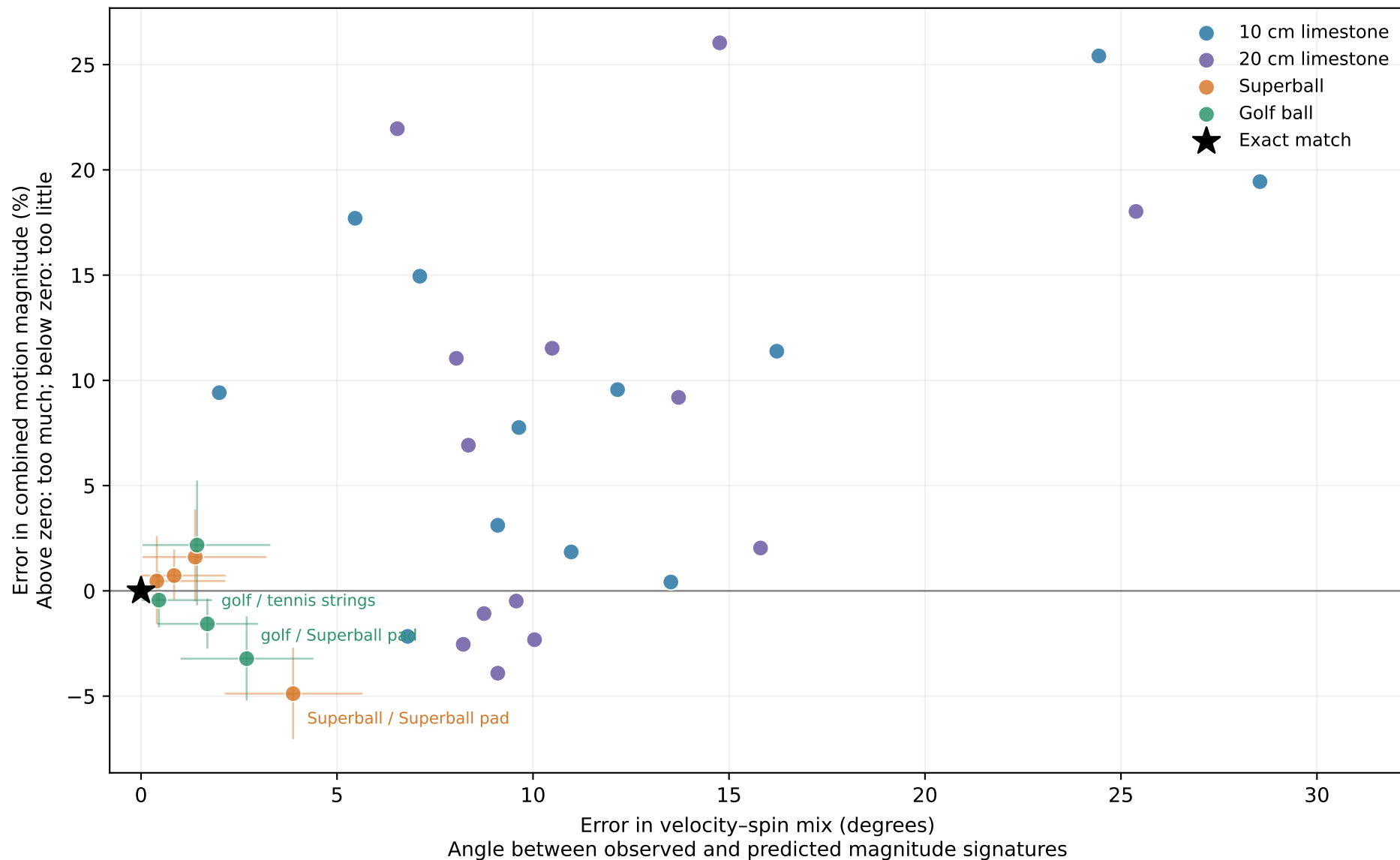


Prediction versus reality — one point per comparison



Signature: ($|v_{\text{normal}}|$, $|v_{\text{tangent}}|$, ell $|\omega|$), with ell = radius. 25 held-out rocks + 8 ball-surface summaries.
 Ball lines: reported input/spin uncertainty envelope, not confidence intervals. Rock uncertainties unavailable.